

The background is a light gray surface with a faint, stylized world map. It is decorated with large, expressive black ink splatters and blotches, primarily concentrated in the corners and along the edges, creating a high-contrast, artistic effect.

Decision tree and Random Forest Algorithms

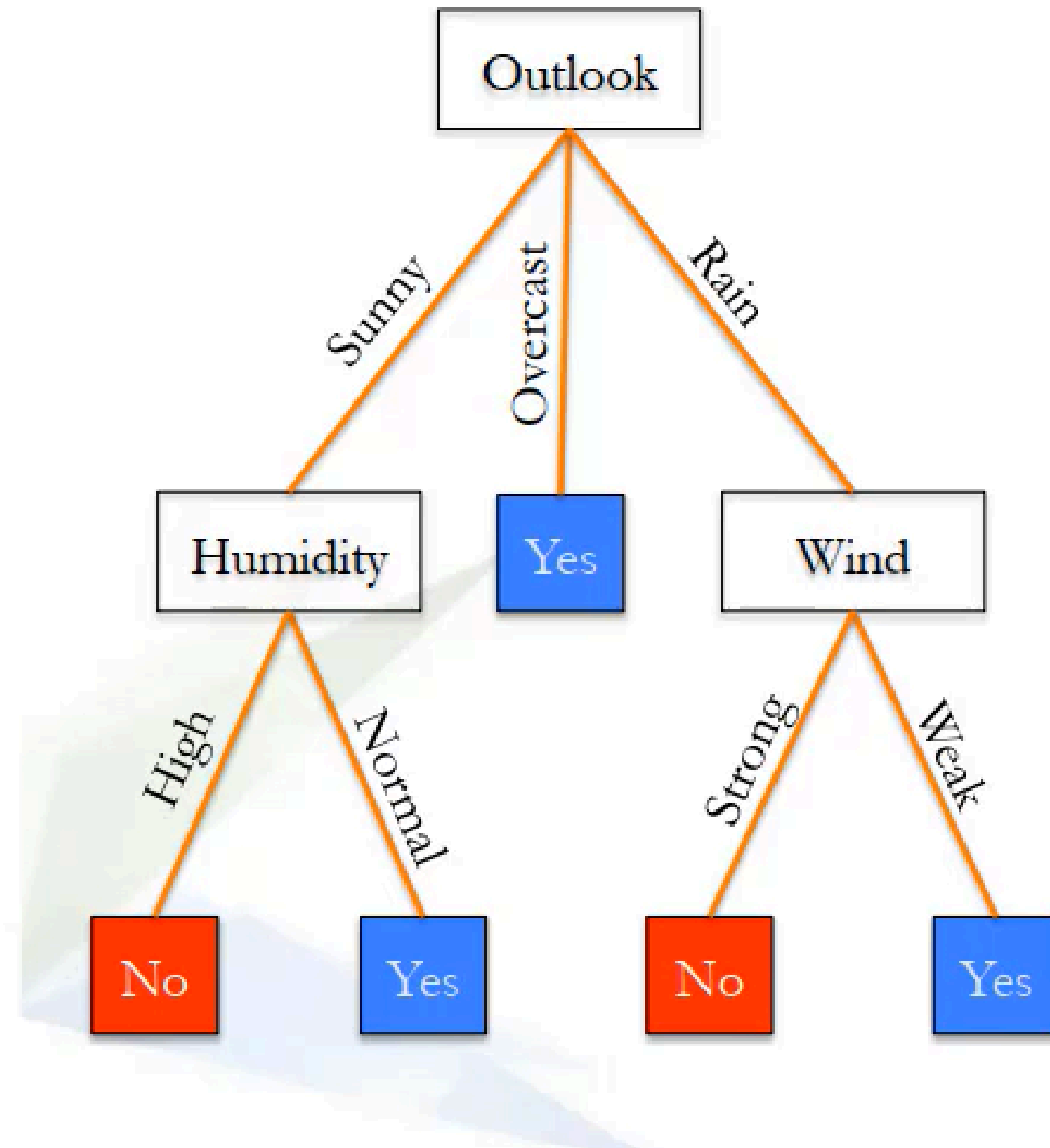
Will Nadal Play Tennis?



Rafael Nadal

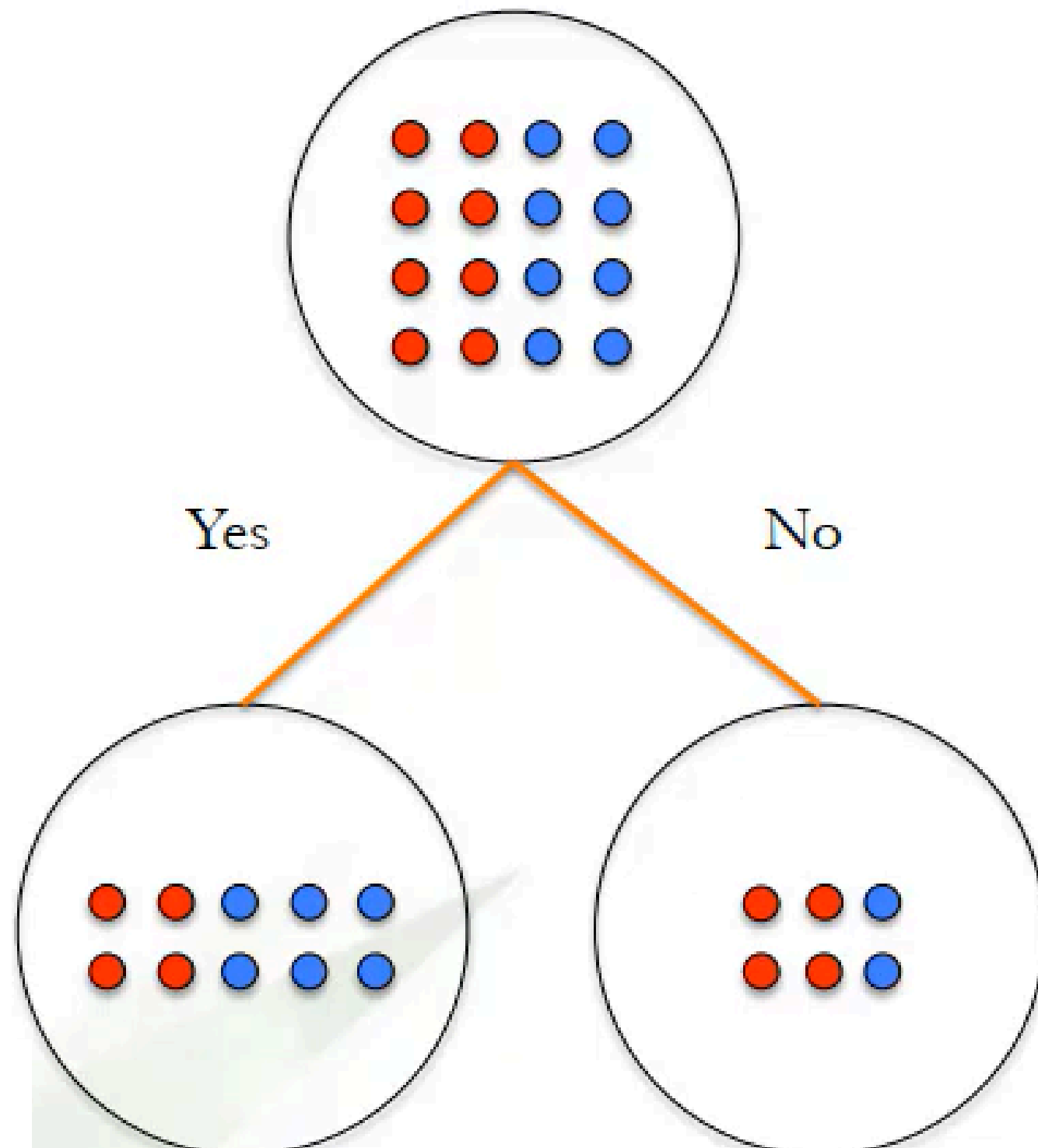
Day	Outlook	Temp	Humidity	Wind	Play Tennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Weak	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Strong	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

Will Nadal Play Tennis?

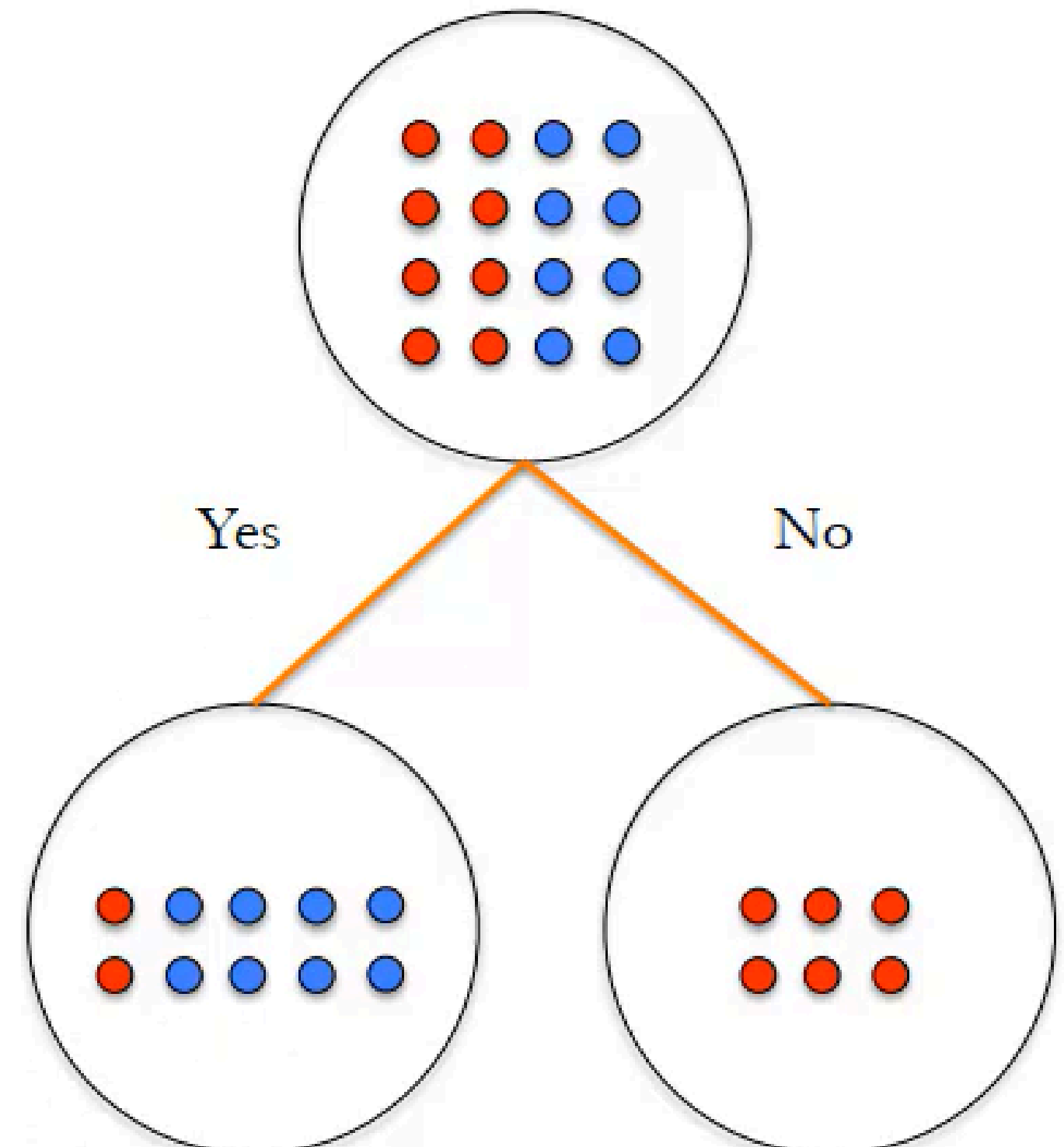


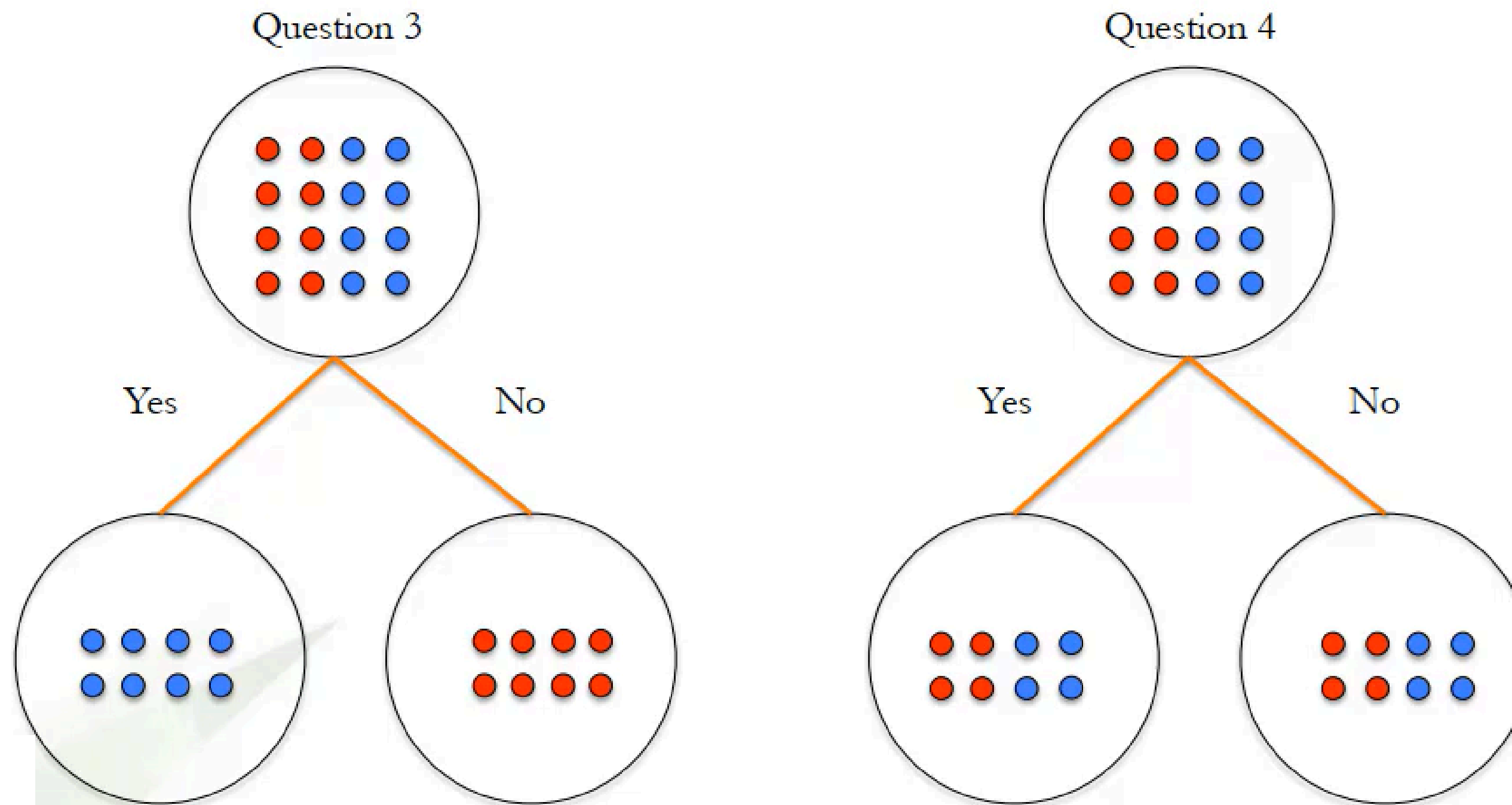
Day	Outlook	Temp	Humidity	Wind	Play Tennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Weak	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Strong	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

Question 1



Question 2



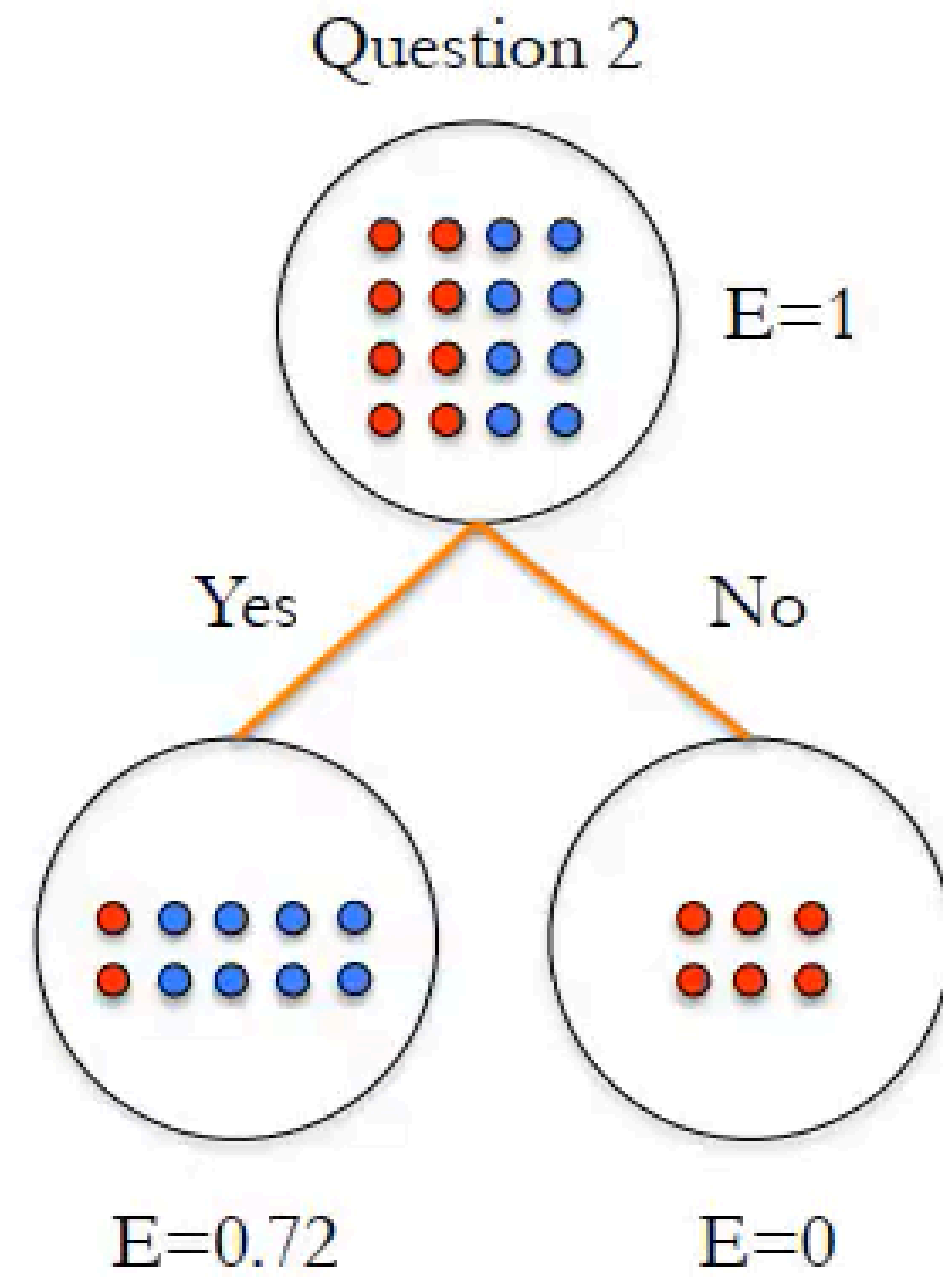
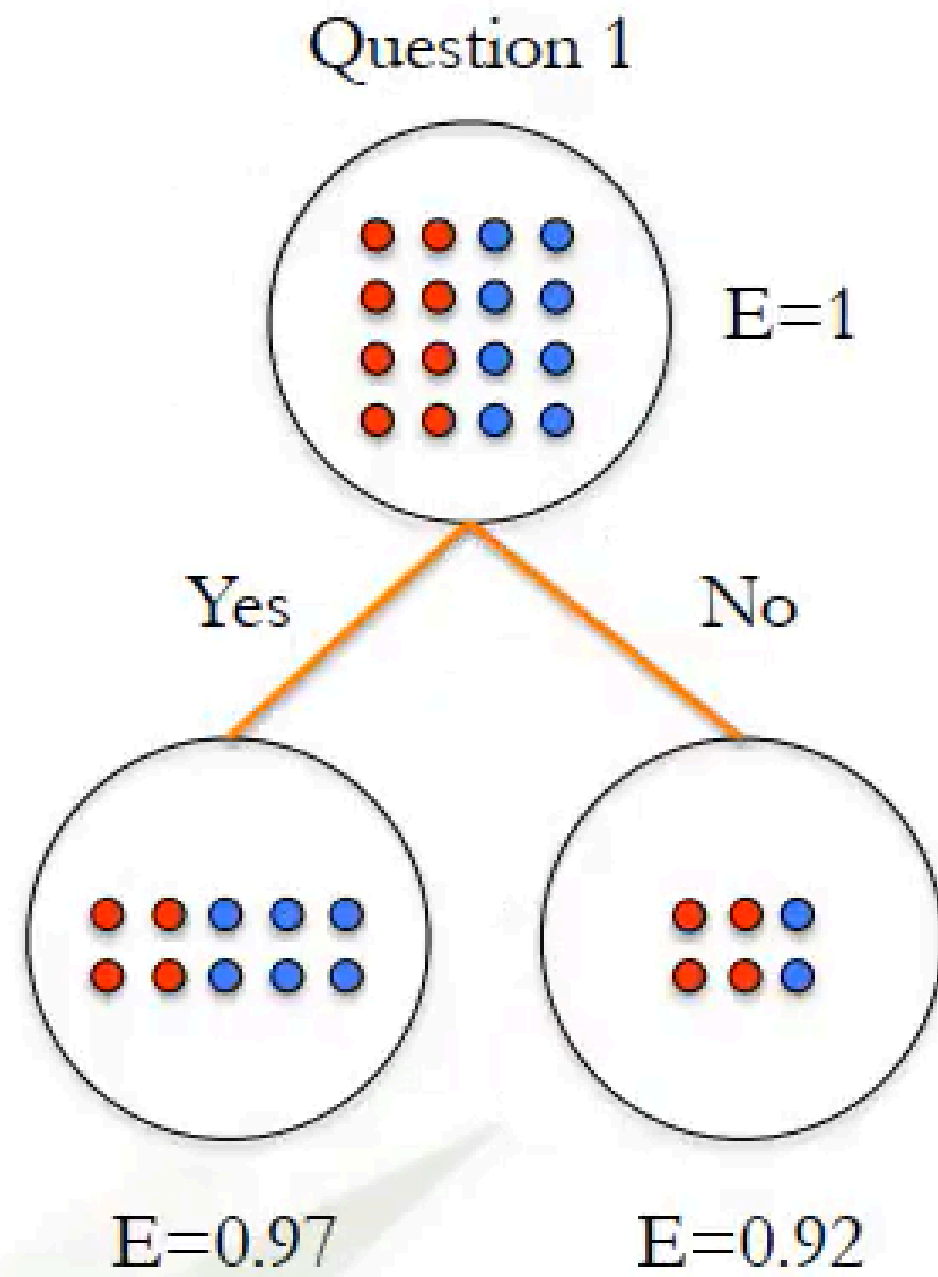


$$E(S) = -p_+ \log(p_+) - p_- \log(p_-)$$



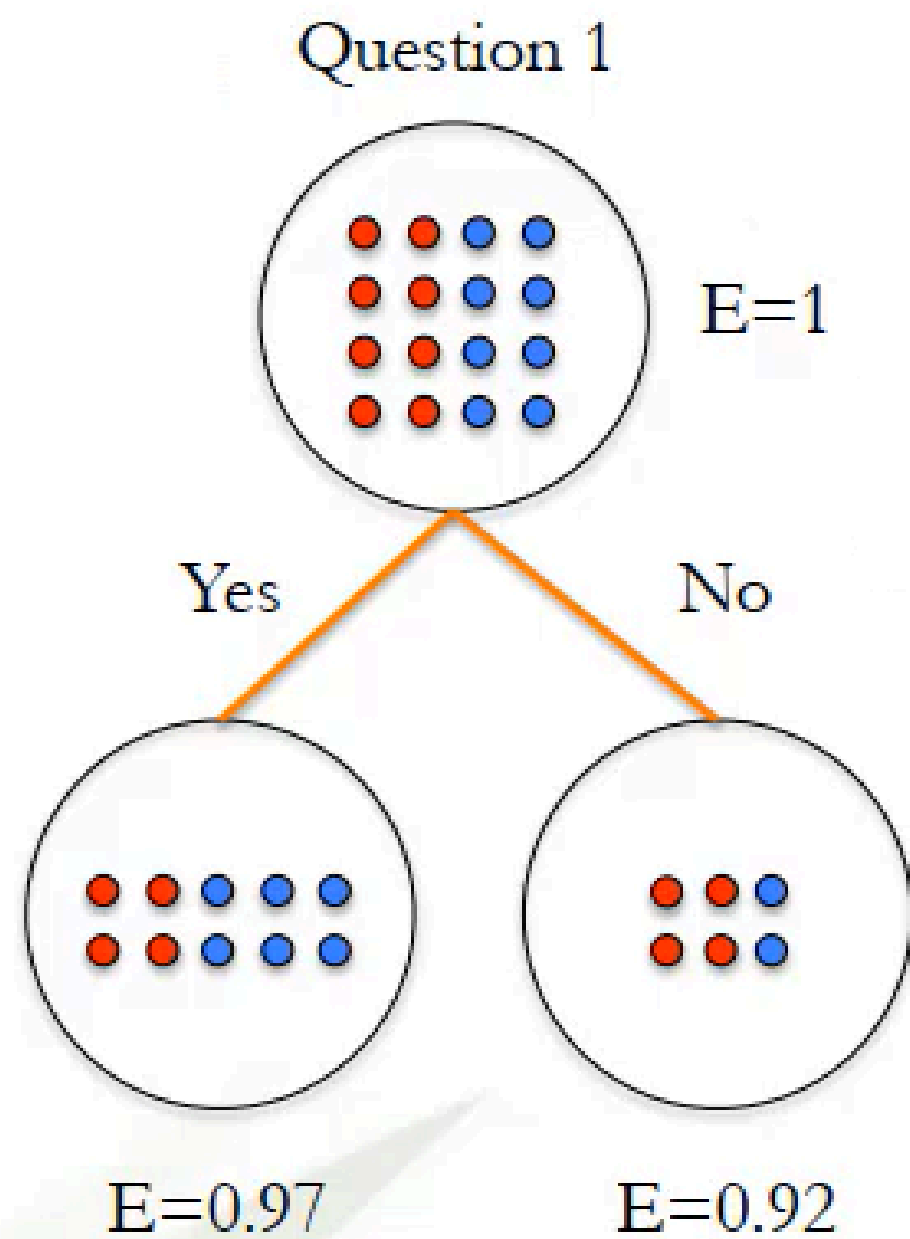
$$y = - \sum_{i=1}^k p_i \log_k(p_i)$$

$$y = - \underbrace{\left[\left(\frac{1}{10} \right) \log_4 \left(\frac{1}{10} \right) \right]}_{\text{Red}} - \underbrace{\left[\left(\frac{3}{10} \right) \log_4 \left(\frac{3}{10} \right) \right]}_{\text{Green}} - \underbrace{\left[\left(\frac{2}{10} \right) \log_4 \left(\frac{2}{10} \right) \right]}_{\text{Blue}} - \underbrace{\left[\left(\frac{4}{10} \right) \log_4 \left(\frac{4}{10} \right) \right]}_{\text{Yellow}}$$



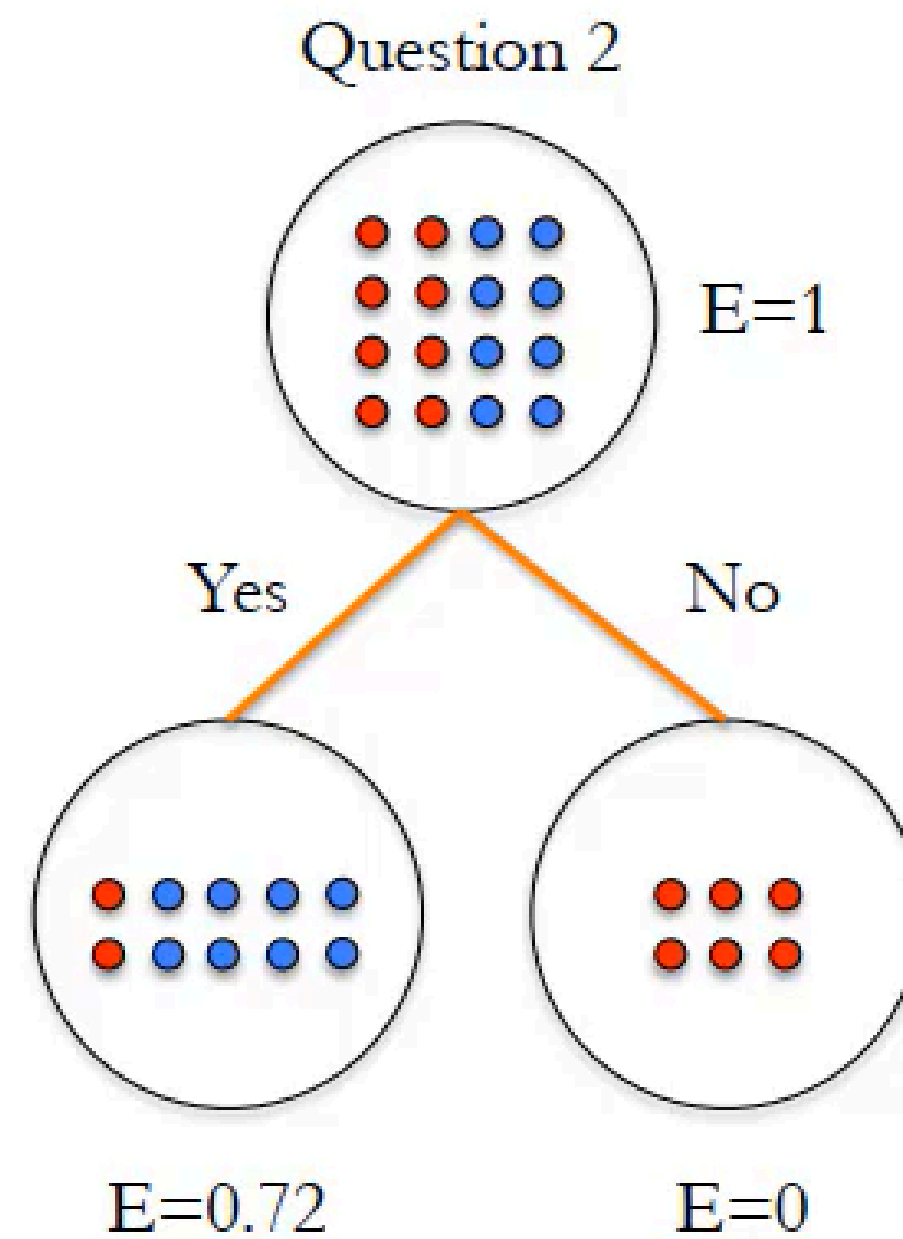
Information Gain

$$G(S, Q) = E(S) - \sum_{i=1}^k p_i E(S, Q_i)$$



$$G(S, Q_1) = 1 - \left(\frac{10}{16}\right) \times 0.97 - \left(\frac{6}{16}\right) \times 0.92$$

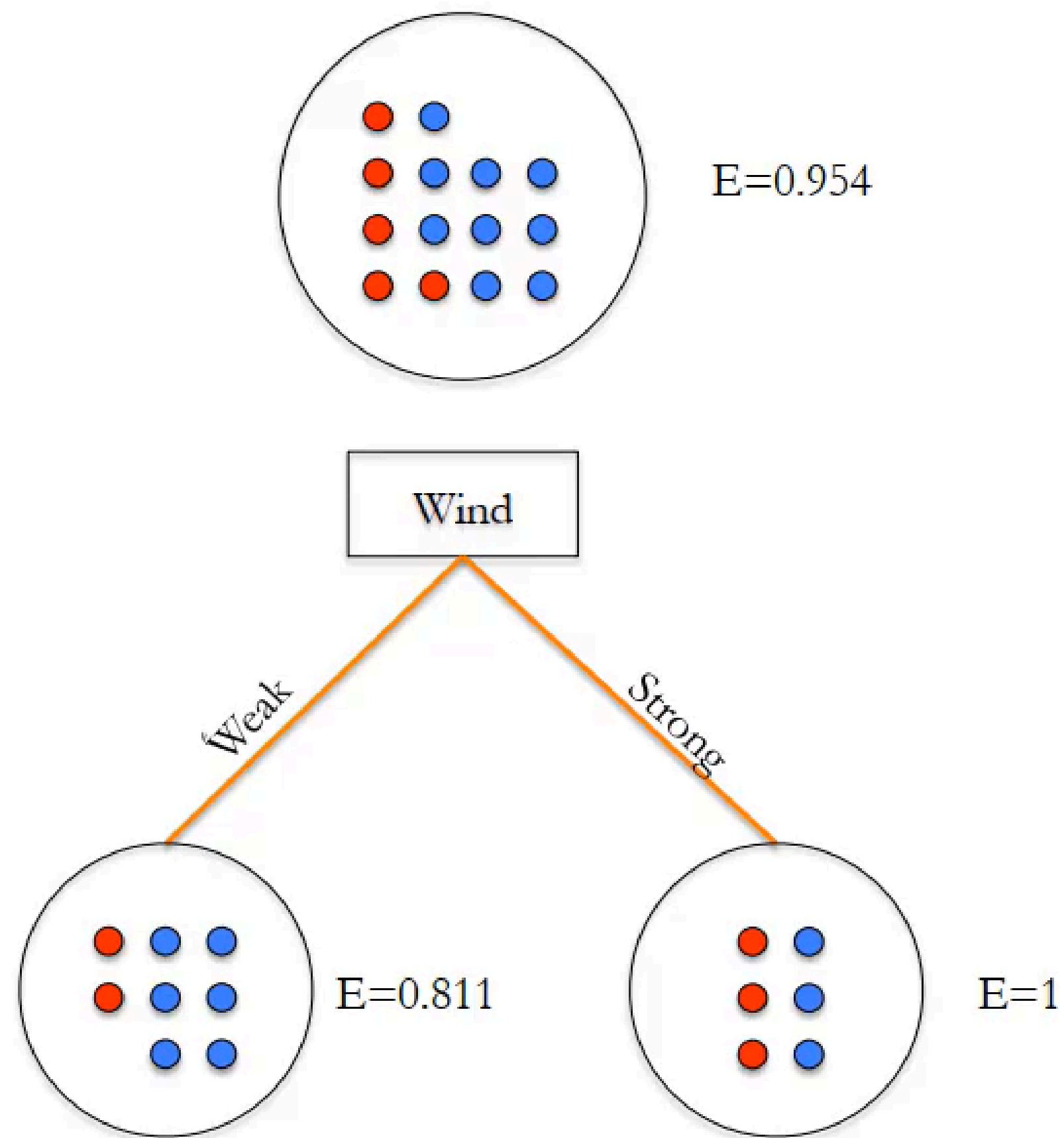
$$G(S, Q_1) = 0.049$$



$$G(S, Q_2) = 1 - \left(\frac{10}{16}\right) \times 0.72 - \left(\frac{6}{16}\right) \times 0$$

$$G(S, Q_2) = 0.55$$

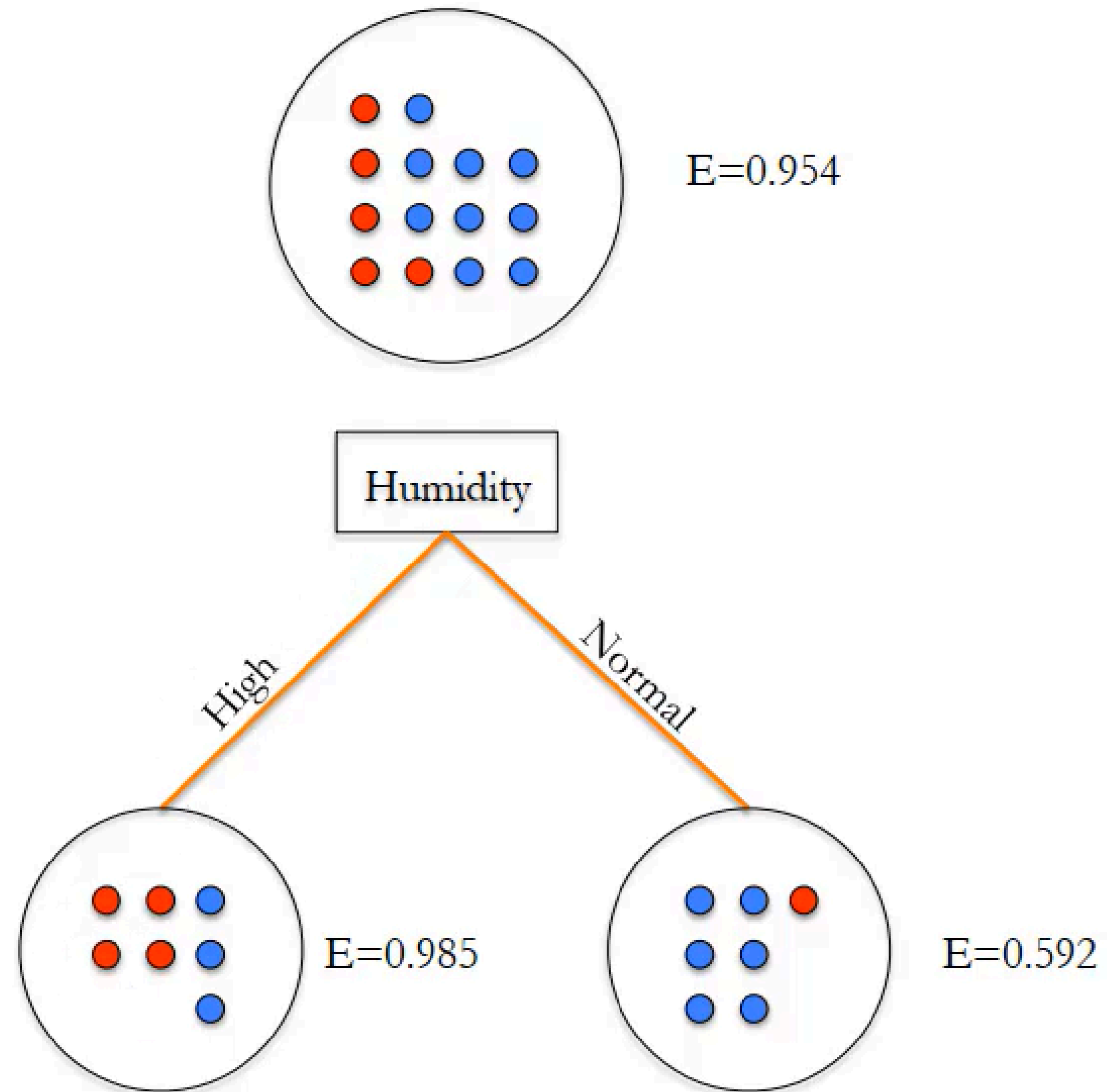
Day	Outlook	Temp	Humidity	Wind	Play Tennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Weak	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Strong	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No



$$G(S, Wind) = 0.954 - \frac{8}{14}0.811 - \frac{6}{14}1$$

$$G(S, Wind) = 0.048$$

$$G(S, Wind) = 0.048$$

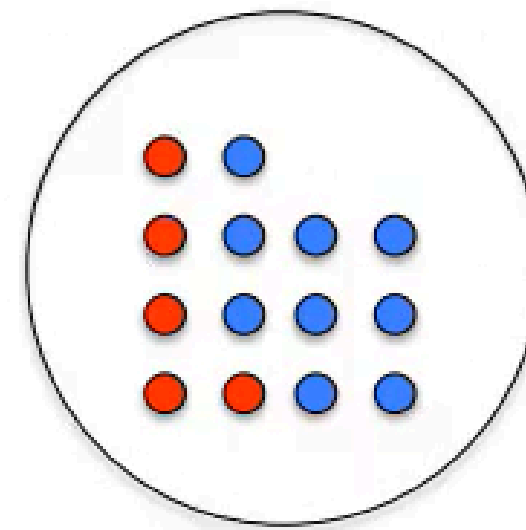


$$G(S, Humidity) = 0.954 - \frac{7}{14}0.985 - \frac{7}{14}0.592$$

$$G(S, Humidity) = 0.151$$

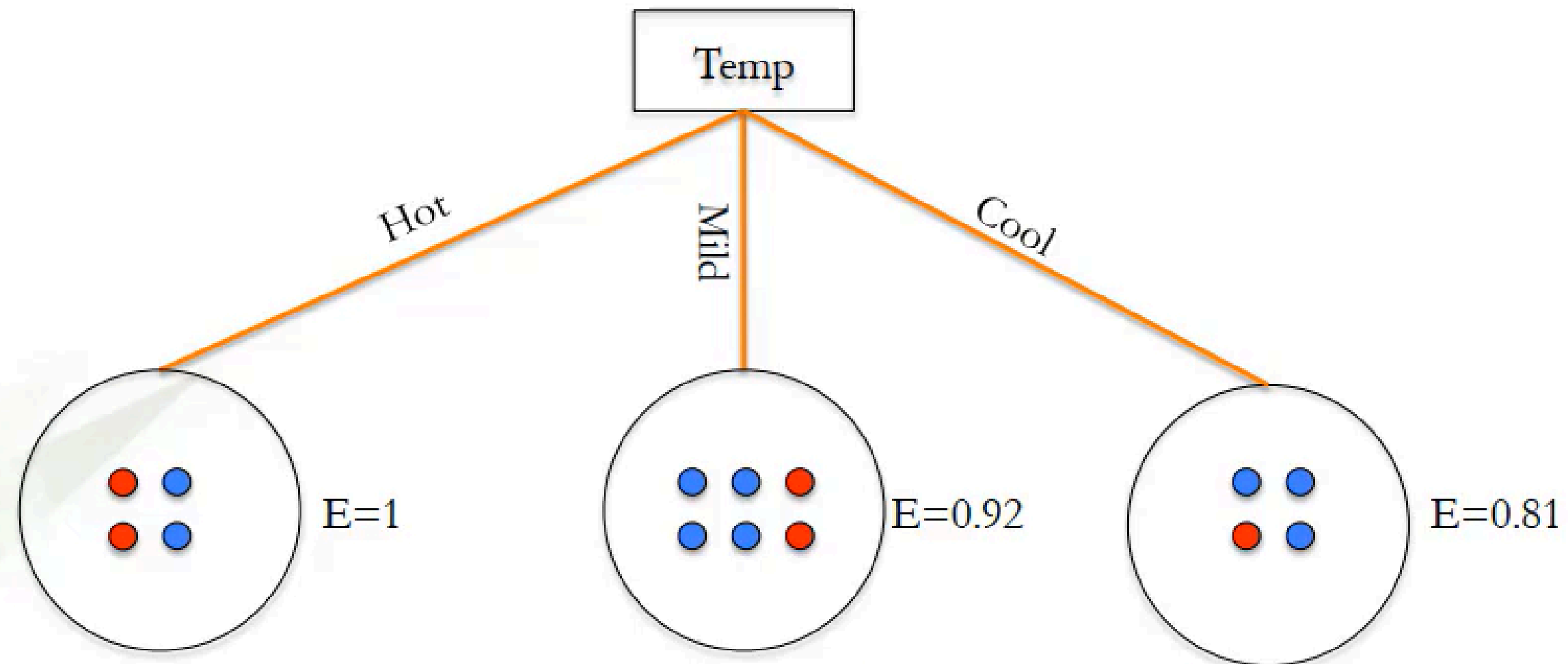
$$G(S, Wind) = 0.048$$

$$G(S, Humidity) = 0.151$$



$$E=0.954$$

30



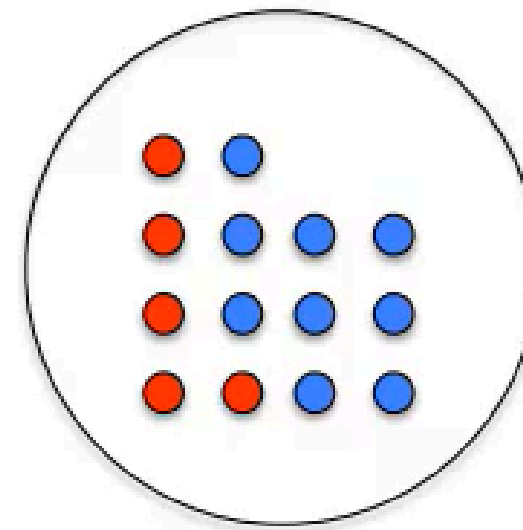
$$G(S, Temp) = 0.954 - \frac{4}{14}1 - \frac{6}{14}0.92 - \frac{4}{14}0.81$$

$$G(S, Temp) = 0.042$$

$$G(S, Wind) = 0.048$$

$$G(S, Humidity) = 0.151$$

$$G(S, Temp) = 0.042$$



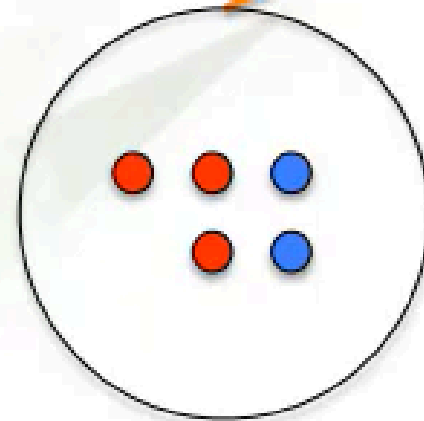
$$E=0.954$$

Outlook

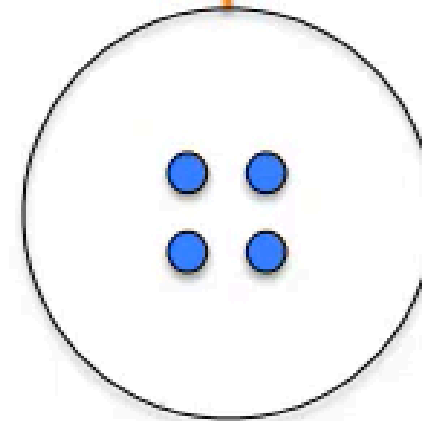
Sunny

Overcast

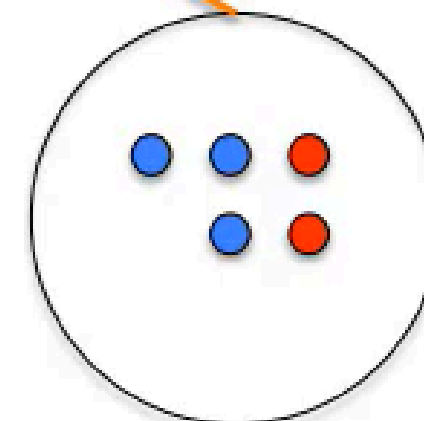
Rain



$$E=0.971$$



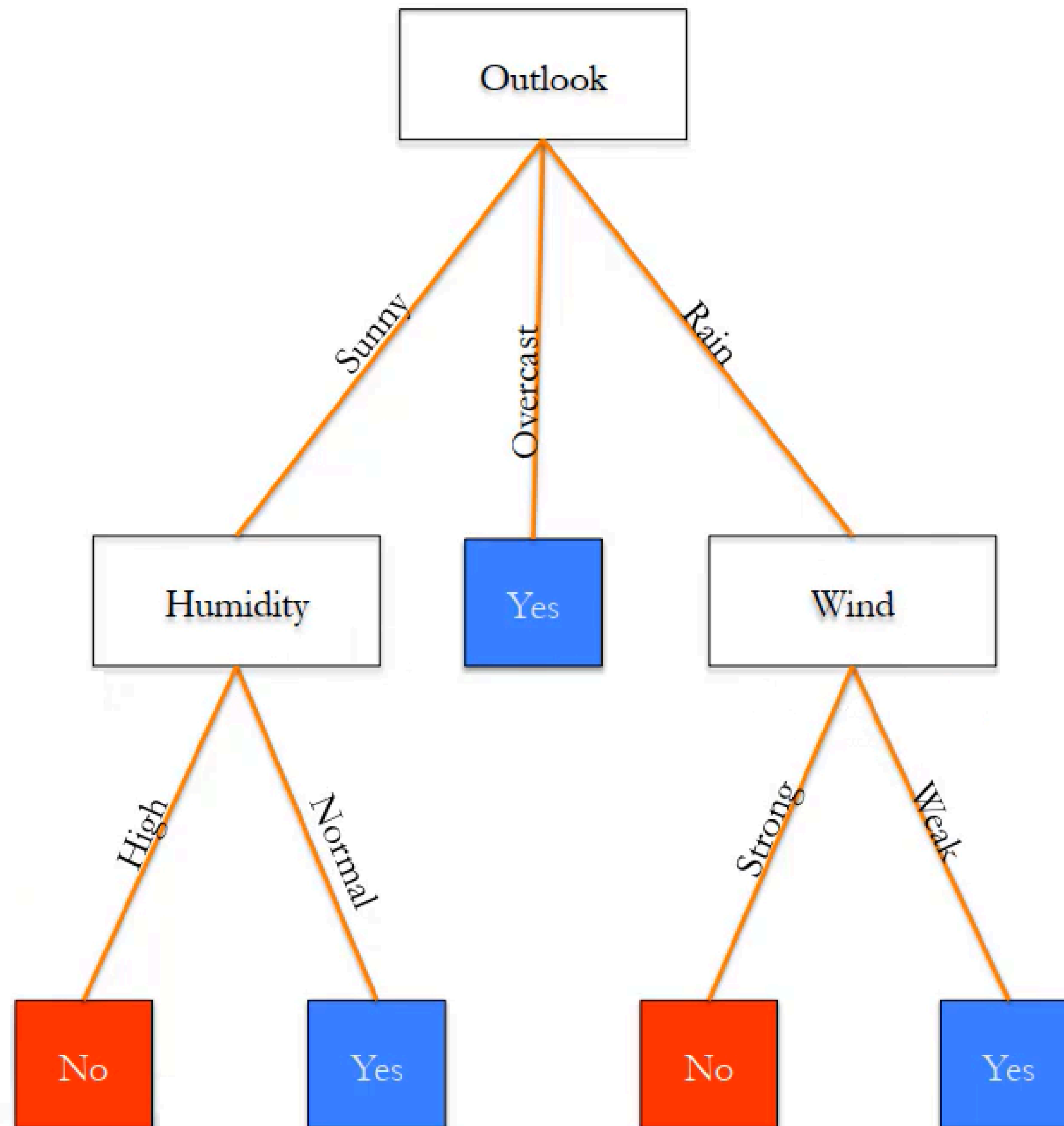
$$E=0$$



$$E=0.971$$

$$G(S, Outlook) = 0.954 - \frac{5}{14}0.971 - \frac{4}{14}0 - \frac{5}{14}0.971$$

$$G(S, Outlook) = 0.247$$



Sunny Outlook data

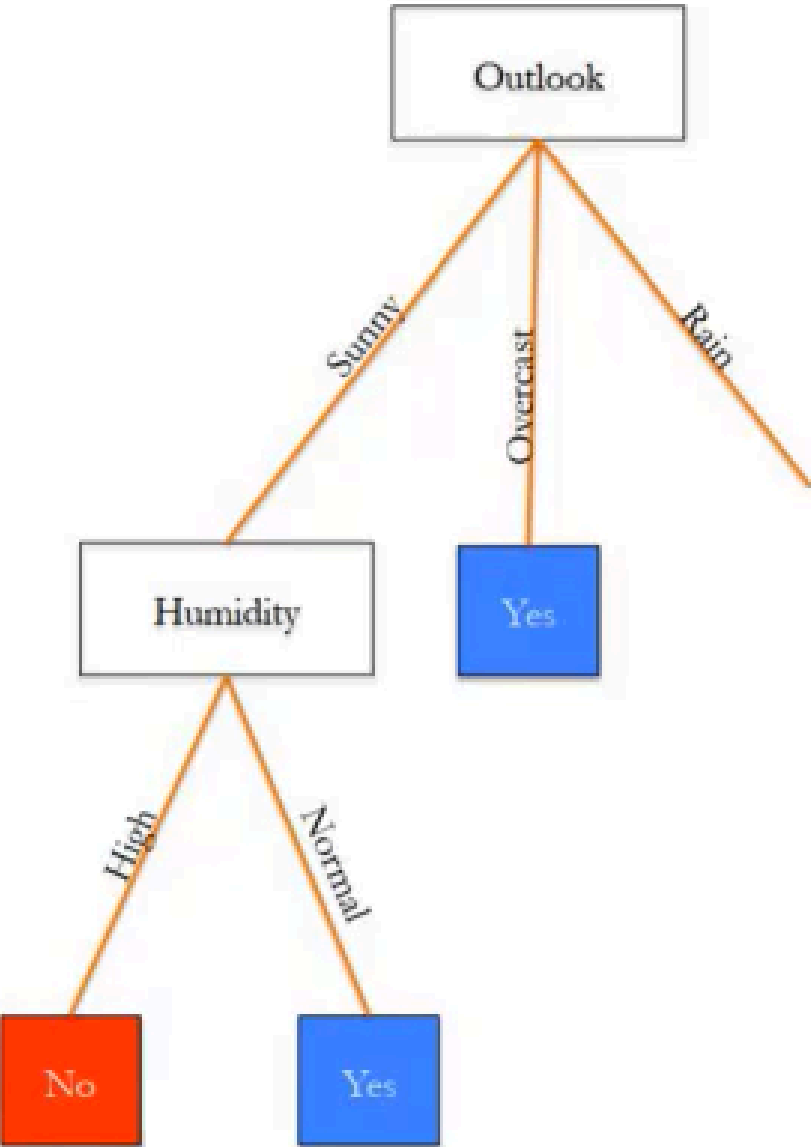
Outlook	Temperature	Humidity	Windy	PlayTennis
Sunny	Hot	High	Weak	No
Sunny	Hot	High	Strong	No
Sunny	Mild	High	Weak	No
Sunny	Cool	Normal	Weak	Yes
Sunny	Mild	Normal	Strong	Yes

Rainy Outlook data

Outlook	Temperature	Humidity	Windy	PlayTennis
Rainy	Mild	High	Weak	Yes
Rainy	Cool	Normal	Weak	Yes
Rainy	Cool	Normal	Strong	No
Rainy	Mild	Normal	Weak	Yes
Rainy	Mild	High	Strong	No

Attributes Gain Data

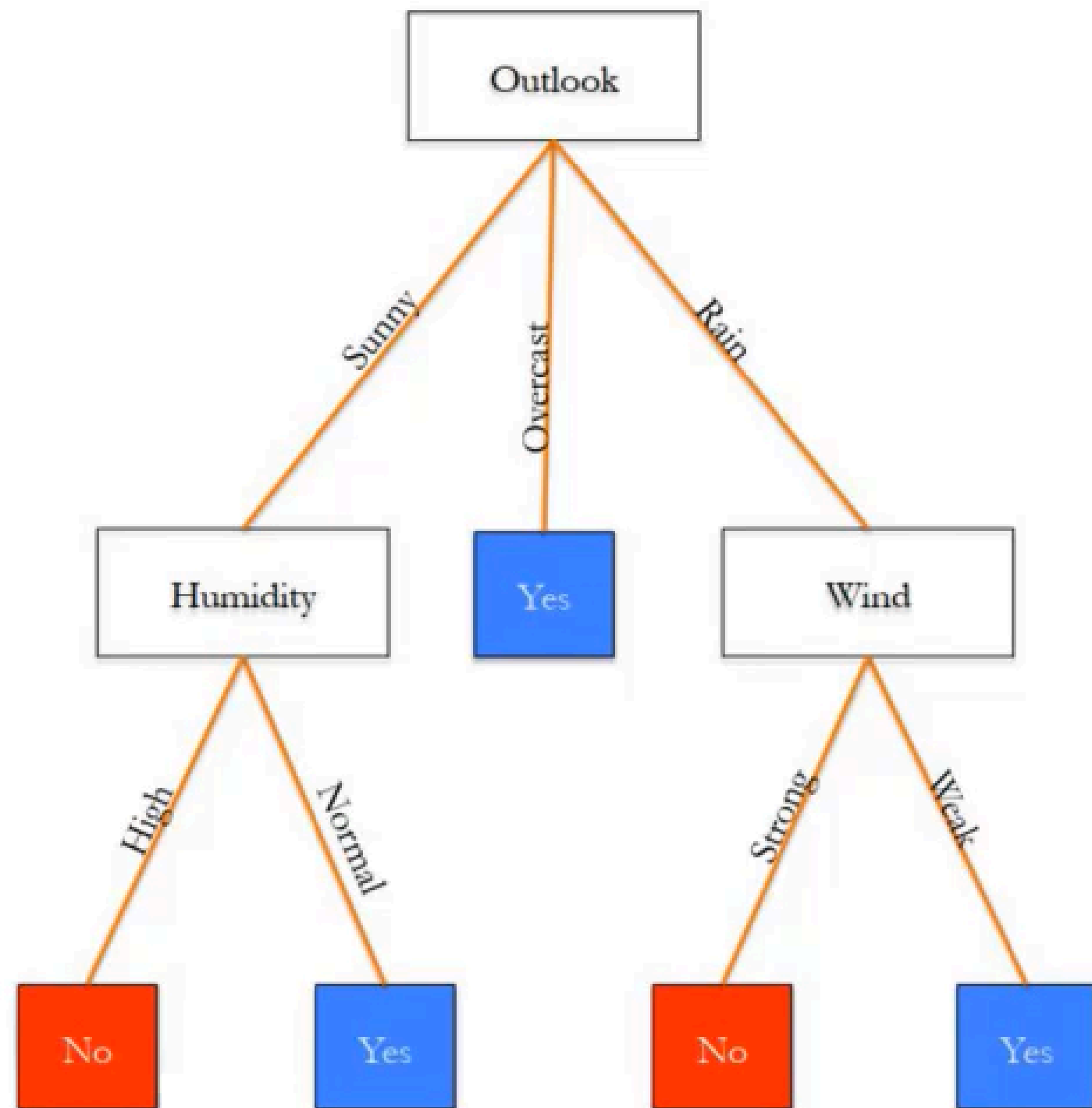
The next node is the humidity



Attributes	Gain
Temperature	0.571
Humidity	0.971
Windy	0.020

Outlook	Humidity	PlayTennis
Sunny	High	No
Sunny	High	No
Sunny	High	No
Sunny	Normal	Yes
Sunny	Normal	Yes

Highest Gain Attributes Data

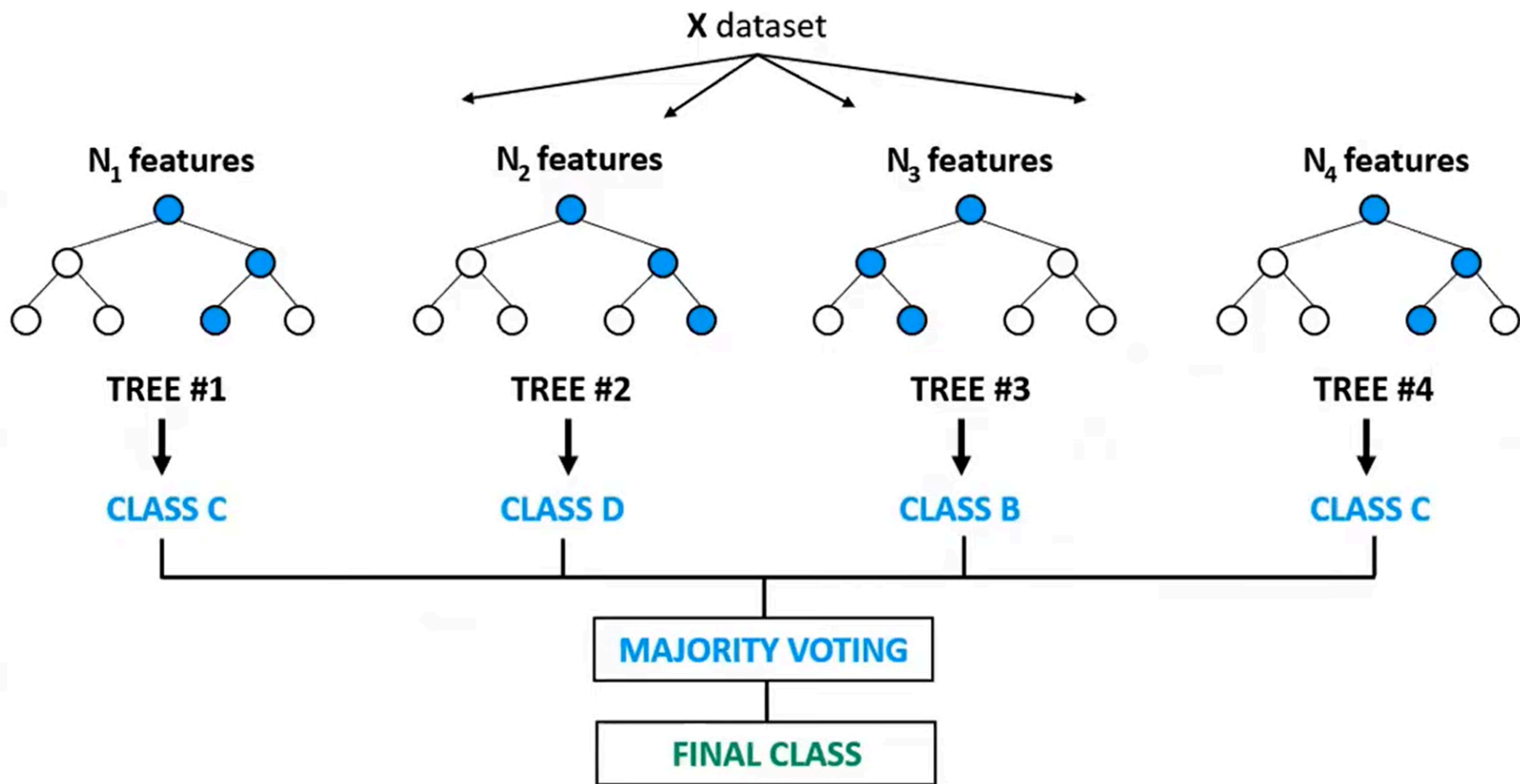


Attributes	Gain
Humidity	0.020
Windy	0.971
Temperature	0.020

Outlook	Windy	PlayTennis
Rainy	Strong	No
Rainy	Strong	No
Rainy	Weak	Yes
Rainy	Weak	Yes
Rainy	Weak	Yes

Random Forest

Day	Outlook	Temp	Humidity	Wind	Play Tennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
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13	Overcast	Hot	Normal	Weak	Yes
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Thanks

Any Questions ?

Eng Ali Raafat